

NAKURU, Kenya

WMO: 637140

Lat: 0.2710S

Lon: 36.1042E

Elev: 6237

StdP: 11.67

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.2	51.9	30.6	31.2	79.6	34.3	36.6	77.7	18.9	68.8	17.3	69.6	0.5	270	0.326

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.0	84.4	56.9	82.8	57.1	81.3	57.5	64.3	73.5	63.5	72.7	62.9	72.0	11.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	103.3	66.6	60.9	101.0	65.8	60.3	98.7	65.3	33.1	73.6	32.4	72.9	31.9	72.3	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.1	16.5	13.1	44.9	87.8	5.6	1.6	40.8	89.0	37.5	89.9	34.4	90.8	30.3	91.9		
(5)			42.0	68.5	3.2	2.1	39.7	69.9	37.8	71.1	36.0	72.3	33.6	73.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.5	67.3	69.0	68.9	67.8	66.4	65.8	64.8	65.0	65.7	65.9	65.1	66.1
	DBStd	2.50	2.19	2.29	2.58	2.33	1.73	1.83	1.86	1.86	2.00	2.10	1.97	2.13
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	141	6	1	2	3	6	12	26	22	14	14	22	13
	CDD50	6010	536	533	586	534	508	473	460	466	471	493	452	498
	CDD65	675	77	114	123	86	48	35	20	23	36	42	25	46
	CDH74	4905	808	1159	1111	497	174	114	71	102	254	222	109	284
	CDH80	637	80	217	237	66	3	2	1	2	13	8	2	6
Wind	WSAvg	3.5	4.4	5.2	4.5	2.9	2.9	2.7	2.9	3.2	3.1	3.2	2.9	3.9
Precipitation	PrecAvg	37.0	1.5	1.4	3.0	5.3	4.3	3.2	3.5	3.9	3.5	3.5	3.1	2.3
	PrecMax	55.6	6.2	5.9	7.7	15.4	10.0	6.5	6.9	8.0	8.7	7.6	8.4	7.5
	PrecMin	24.2	0.3	0.1	0.0	1.3	1.6	0.8	0.3	1.2	0.9	1.1	0.4	0.0
	PrecStd	7.2	1.3	1.5	1.7	3.1	1.9	1.5	1.7	1.6	1.8	1.6	1.7	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.9	86.0	86.4	84.7	80.3	79.2	78.4	79.0	81.5	80.9	79.0	80.9
		MCWB	56.5	56.2	56.7	58.4	60.5	61.2	60.0	59.1	58.4	58.9	59.5	57.9
	2%	DB	82.2	84.3	84.6	81.8	77.7	77.0	75.8	76.6	79.1	78.6	76.9	78.9
		MCWB	56.3	56.3	57.0	59.1	60.9	61.1	60.3	60.0	59.3	59.7	59.8	58.1
	5%	DB	80.6	82.7	82.8	79.4	76.1	75.1	74.1	74.8	77.0	76.7	75.0	77.2
		MCWB	56.7	56.4	57.5	59.8	61.1	61.2	60.2	60.2	59.4	59.9	60.2	58.6
	10%	DB	78.8	81.0	80.8	76.9	74.1	73.3	72.2	72.9	75.0	74.5	73.1	75.5
		MCWB	57.0	56.7	57.8	60.5	61.2	60.9	59.9	59.9	59.7	59.9	60.3	58.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.2	64.3	63.7	64.8	65.2	64.7	63.7	63.5	63.6	64.1	63.9	63.9
		MCDB	72.6	75.1	74.9	74.5	74.0	73.8	73.6	73.6	74.5	73.5	71.7	71.2
	2%	WB	61.8	62.0	62.5	63.7	63.8	63.4	62.5	62.2	62.2	62.7	62.8	62.8
		MCDB	71.2	71.9	73.8	72.7	72.8	72.8	72.1	72.6	73.0	71.9	70.5	70.6
	5%	WB	60.9	60.7	61.6	62.9	62.9	62.6	61.5	61.3	61.3	61.8	62.0	61.9
		MCDB	70.6	71.5	72.6	71.8	71.5	72.1	70.9	71.2	71.8	70.6	69.5	69.8
	10%	WB	60.0	59.5	60.8	62.1	62.1	61.7	60.6	60.5	60.5	61.0	61.3	61.1
		MCDB	70.6	72.0	71.5	70.6	70.2	70.4	69.6	69.8	70.6	69.5	68.6	68.9
Mean Daily Temperature Range	5% DB	MDBR	24.9	26.0	24.4	20.2	18.1	18.1	18.1	19.2	21.5	19.8	17.6	20.6
		MCDBR	28.2	28.6	28.5	24.5	21.1	20.4	20.3	21.4	24.7	23.0	21.0	23.5
		MCWBR	8.4	7.8	7.6	7.0	7.5	7.7	7.9	8.2	8.8	7.9	7.4	7.4
	5% WB	MCDBR	21.6	23.7	21.7	18.2	17.6	17.9	18.2	19.3	21.2	19.2	17.1	17.8
		MCWBR	7.8	8.1	7.3	6.8	7.2	7.7	8.0	8.3	8.7	8.0	7.2	7.3
Clear-Sky Solar Irradiance	taub	0.337	0.359	0.359	0.328	0.330	0.391	0.409	0.383	0.350	0.346	0.337	0.336	
	taud	2.521	2.439	2.469	2.615	2.601	2.376	2.300	2.374	2.469	2.490	2.547	2.542	
	Ebn at Noon	315	308	305	308	300	277	274	287	303	308	312	314	
	Edh at Noon	35	38	37	31	30	37	41	39	36	36	34	34	
All-Sky Solar Radiation	RadAvg	1961	2073	2014	1796	1764	1630	1565	1643	1865	1820	1671	1801	
	RadStd	155	121	152	121	69	99	116	96	81	92	109	132	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page